GG-R-001638 (DSA-DM)
April 9, 1970
(See section 6.3)

INTERIM FEDERAL SPECIFICATION

REMOVER, DENTAL CROWN, SIMPLEX

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal Specification.

1. SCOPE

1.1 Scope. This specification covers a crown remover suitable for use during dental operative procedures.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
QQ-B-626	Brass, Leaded and Nonleaded, Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip).
MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

FSC 6520

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-P-116	Methods of Preservation.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation	E18	Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM Designation	E92	Tests for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 General requirements. The crown removers shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material.

3.2.1 Shaft and points. The shaft and points shall be fabricated of Class 1, Types 1042, 1043, or 1045 carbon steel, except that the sharpened surface of the working ends may be unplated.

3.2.2 Weight, sleeve, and cap. The weight, sleeve, and cap shall be fabricated of chrome-plated brass. The brass shall conform to the chemical requirements for Composition 11 of QQ-B-626. The weight, sleeve, and cap shall pass the boil test specified in GG-I-526.

3.3 Hardness. After proper stress relieving, the shaft and points shall have a Rockwell hardness of C40 to C45.

3.4 Style, design, and dimensions. The crown removers shall be of the type known as "Simplex", and shall be suitable for use during dental operative procedures. The crown removers shall be furnished with one contra-angle point (Type 1) and one right-angle point (Type 2). The points shall be of the cone socket type, and shall be readily interchangeable into the distal end of the shaft. The style, design, and dimensions of the crown removers shall be in accordance with figure 1.

3.4.1 Figure. Figure 1, forming a part of this specification, is a specification control drawing. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the item. No deviation from the prescribed dimensions on tolerances is permissible without prior approval of the contracting officer.

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3.5 Construction. The crown removers shall consist of a cap, shaft, weight, sleeve, and two interchangeable points. The cap shall be knurled, and shall be securely and permanently attached to the proximal end of the shaft. The weight shall move freely along the shaft between the cap and the sleeve (stops). The weight shall weigh 55 ± 3 grams. The threaded, knurled sleeve shall be removable from the distal end of the shaft to allow for the removal of the weight. The cap shall withstand the impact test without failure, when tested as specified in 4.4.2.

3.6 Finish. The points shall have a mirror finish, except for the sharpened working end which may have a fine honed finish. The remainder of the crown remover shall have a mirror or satin finish, except for the knurled portions.

3.7 Corrosion resistance. The shaft and points shall meet the salt-spray test requirements of GG-I-526, and the entire assembled crown remover shall meet the boil test requirements of GG-I-526.

3.8 Identification marking. Crown removers shall be permanently and legibly marked in accordance with GG-I-526. Contra-angle points (Type 1) shall be marked with the number "1", and straight angle points (Type 2) shall be marked with the number "2". Each crown remover and each point shall be permanently and legibly marked with the name or registered trademark of the manufacturer (and contractor, if other than manufacturer). Instruments of foreign manufacture shall be permanently and legibly marked with the name of the country of origin, together with the name or registered trademark of the contractor and manufacturer.

3.9 Workmanship. Workmanship shall be first class throughout. The crown removers shall be free from defects which either detract from their appearance or may impair their serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel and brass, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the material, the lot number, individual test results, and specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of these items or lots of items to be offered are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one crown remover.

Table I. Examination Criteria

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II, except that the acceptance number shall be zero for all sample sizes. The unit of product for inspection purposes shall be one crown remover.

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Table II. Tests

Component	Characteristic	Requirements	Test procedure	Inspection level
Shaft and points	Hardness	3.3	GG-I-526	S-2
	Plating thickness	3.2.1	GG-I-526	S-2
	Salt spray	3.2.1	GG-I-526	S-2
Entire assembled instrument	Boil test	3.2.1 and 3.2.2	GG-I-526	S-2
	Impact test	3.5	4.4.2	S-2

4.2.3 Raw materials. A sufficient quantity of each lot or metal used in the construction of the crown removers shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

4.3 Examination. The crown removers shall be examined to determine compliance with the requirements contained in this specification. Examination shall be conducted as specified in 4.2.1. Any crown remover in the sample containing one or more defects shall be rejected, and if the number of defective crown removers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The crown removers shall be examined for defects including, but not limited to, those listed in table III.

Table III. Classification of Defects

Category	Defect
Major A:	
101	Style and design not as specified.
102	Crown remover working surfaces not free from burrs.
103	Crown remover not free from cracks.
104	Finish of points not as specified.
105	Construction not as specified.
Major B:	
151	Dimensions of operative ends of points not as shown on applicable figure.

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Table III. Classification of Defects (Con't)

Minor A:

201	Identification marking not as specified.
202	Finish of cap, sleeve, shaft, and weight not as specified.
203	Finish not free of coarse grind marks.
204	Crown remover not free from dirt, polishing compound, or foreign matter.
205	Packaging and packing not as specified.
206	Marking on packaging and pack not as specified.

Minor B:

251	Dimensions (other than 151) not as specified.

4.4 Test procedures.

4.4.1 Hardness. The hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E92 (Vickers).

4.4.2 Impact test. The number 2 (straight angle) point shall be securely threaded in the socket of the crown remover. The crown remover shall be securely fixed at a point around the threaded knurled sleeve, or above, in such a manner that the fixation will not interfere with the sliding weight as it moves vertically upward on the shaft, from cap to threaded knurled sleeve. A 20 ℄ 1 pound weight shall be secured to the sliding weight so as not to interfere with the free movement of the sliding weight. The sliding weight, with its added weight, shall be pushed up the shaft until the sliding weight makes contact with the bottom of the threaded knurled sleeve, and then it shall be released. This impact test shall be performed five times, and then the cap shall be examined for evidence of movement or loosening of cap.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One remover, as specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	144 units

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be over-packed in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IA-15.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with either adhesive (conforming to MMM-A-250) or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Specification coverage. This document covers the requirements for FSN 6520-551-6150, Remover, Dental Crown, Simplex.

6.3 Supersession data. This document supersedes Military Specification MIL-R-2208A dated 15 June 1961.

6.4 Specification changes. Because of the extensive changes to this document, the margins have not been annotated with asterisks to indicate where changes from the previous issue were made.

6.5 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
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Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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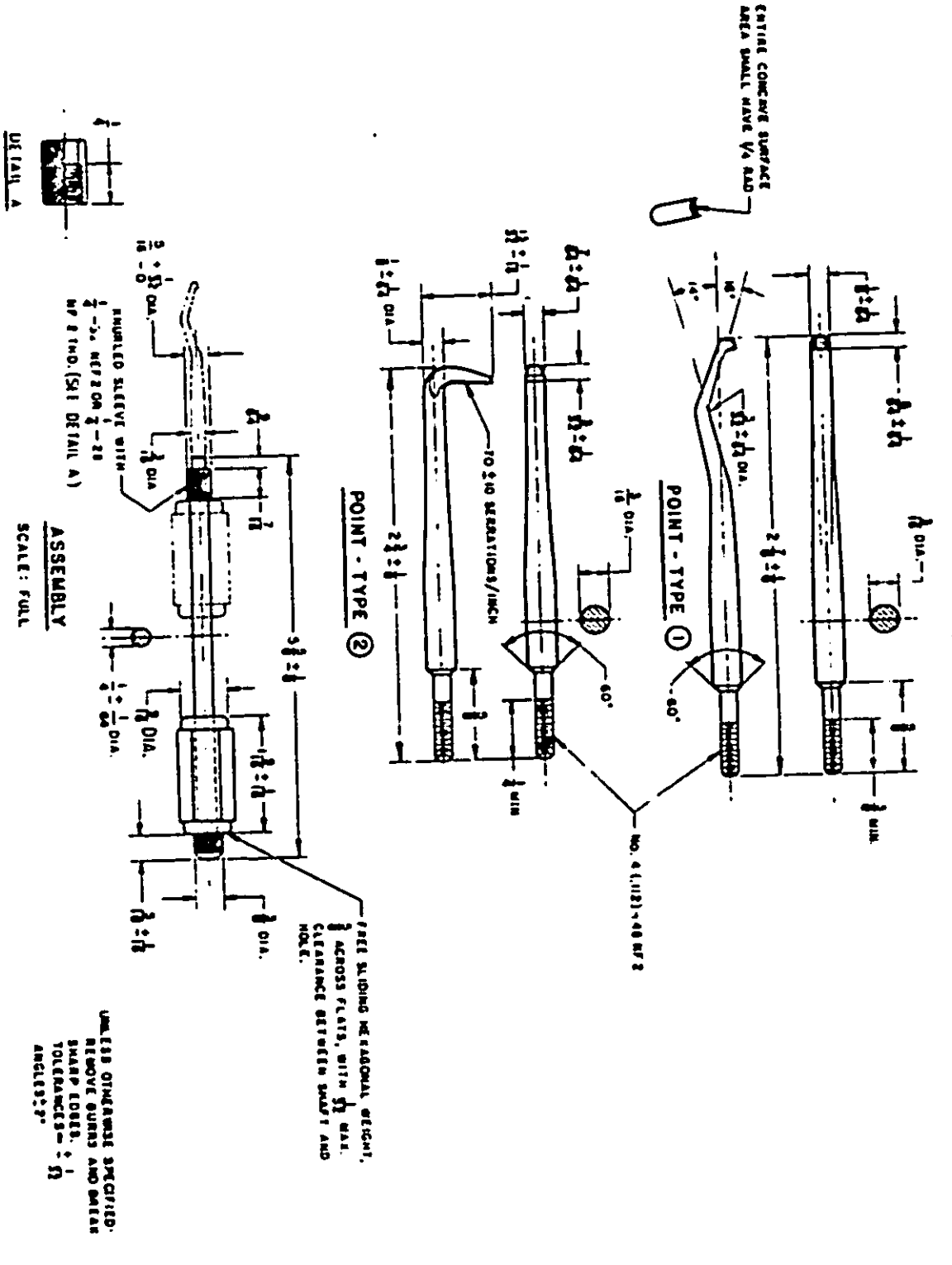


FIG. 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. *Fold on lines on reverse side, staple in corner,* and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

OG-R-001638(DSA-DM) PSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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